AI-Based Crop Recommendation for Farmers Using Deep Learning

Rashid Amir, Ayush Mishra, Praveen Yadav
Department of Computer Science and Engineering
Lovely Professional University
Email: rashid.amir0414@gmail.com, mishraayush0415@gmail.com,
praveenyadav2155@gmail.com.

***Abstract*—The agricultural industry is crucial to numerous developing nations; however, many farmers depend on conventional knowledge rather than scientific evidence for selecting crops. This practice yields lower yields and results in ineffective resource utilization. In this study, we introduce an AI-driven deep learning framework for crop recommendations that assesses soil nutrients (N, P, K), environmental factors (temperature, humidity, rainfall), and soil pH to determine the most appropriate crop for a particular area. We develop three different models: (1) a Multi-Layer Perceptron (MLP), (2) a Transformer-based model (TabTransformer), and (3) TabNet, which is a deep learning model based on sequential attention. These models are trained using a publicly available dataset consisting of 2,200 entries and 22 distinct crop types. The evaluation metrics utilized include accuracy, precision, recall, F1-score, and calibration error. Among these, the TabTransformer recorded the highest accuracy of 98.9%, surpassing the performance of the other architectures. Visualizations such as histograms, confusion matrices, and feature importance plots enhance the interpretability of the results. The deep learning framework we propose improves the decision-making process for farmers, promoting sustainable and profitable agricultural practices.**

***Index Terms*—Precision Agriculture, Deep Learning, TabTransformer, TabNet, Crop Recommendation, Explainable AI, Neural Networks.**

I. Introduction

Agriculture continues to be a foundational element of national economies, especially in developing countries such as India. Even with advancements in technology, many farmers depend on traditional intuition for choosing crops, which often results in unsuitable crop-soil pairings and diminished yields. Utilizing data-driven decision-making through artificial intelligence (AI) can enhance crop selection by examining soil nutrients, weather patterns, and rainfall statistics.

Unlike classical machine learning (ML), deep learning (DL) algorithms can independently discover nonlinear relationships among various agro-environmental factors. Advanced models like **TabTransformer** and **TabNet** excel with tabular data, enabling the model to emphasize feature interactions using attention mechanisms and sequential learning.

This research introduces a crop recommendation framework based on deep learning that employs environmental and soil data. The key contributions of this paper include:

- An extensive deep learning architecture that integrates MLP, TabTransformer, and TabNet models.
- A thorough examination of feature significance, accuracy, interpretability, and calibration.
- The incorporation of explainable AI techniques (SHAP, Integrated Gradients) to enhance model transparency.

II. Literature Review

During the past ten years, many researchers have utilized machine learning and AI for predicting crops, estimating yields, and advancing smart agriculture. The next section assesses fifteen academic studies that have played a role in the creation of intelligent crop recommendation systems.

A. 1) ML-based Crop Recommendation Systems

Apat et al. \cite{apat} created a machine learning-driven crop suggestion system by employing Random Forest and SVM on datasets related to soil nutrients. They attained an accuracy rate of 99.2\

B. 2) Sustainable Agriculture via Smart Decision Systems

Aarthi et al. [2] utilized KNN and Decision Tree models to predict sustainable agriculture outcomes. Their approach suggested crop choices based on environmental factors and soil characteristics, emphasizing environmentally friendly crop rotations.

C. 3) Deep Learning in Smart Agriculture

Bakr et al. [3] utilized Deep Neural Networks (DNNs) and Recurrent Neural Networks (RNNs) to predict crop yields, achieving an accuracy of 99.5% and surpassing traditional machine learning models.

D. 4) Explainable AI for Agriculture

Ahamed [4] emphasized explainable AI (XAI) to make model decisions transparent for farmers. SHAP and LIME were employed to visualize how soil and rainfall affected predictions.

E. 5) Ensemble Learning for Crop Recommendation

Kaur and Singh [5] used Gradient Boosted Decision Trees (XGBoost) for regional crop classification, improving F1-score and achieving robust performance across multiple soil regions.

F. 6) IoT and Deep Learning Integration

Reddy et al. [6] proposed an IoT-based smart advisory system where soil sensors transmitted pH, moisture, and temperature to a neural model deployed at the farm edge.

G. 7) Crop Suitability under Climate Change

Hussain and Ali [7] studied crop yield under changing rainfall and temperature conditions using Random Forest. Their model improved adaptive crop planning for drought-prone zones.

H. 8) TabTransformer for Tabular Data

Huang et al. [8] introduced the TabTransformer model, using self-attention for tabular features, enabling context-aware embeddings that inspired this paper's deep learning framework.

I. 9) TabNet: Interpretable Deep Learning for Structured Data

Arik and Pfister [9] proposed TabNet, an interpretable neural model using sequential attention masks, which allows explanation of which features (e.g., pH or rainfall) influence predictions most.

J. 10) Transfer Learning for Crop Yield Prediction

Patel and Kumar [10] combined convolutional and dense networks using transfer learning for yield prediction based on satellite images, achieving scalable results for large regions.

K. 11) Feature Engineering for Soil Fertility

Subramanian et al. [11] applied hybrid DL and feature engineering for soil fertility classification, suggesting fertilizer-crop combinations, improving accuracy to 97.8%.

L. 12) Rainfall-Based Crop Selection

Rai and Singh [12] developed an SVM-based model using rainfall data and humidity to predict suitable crops, concluding rainfall is the dominant factor for tropical crop performance.

M. 13) Sustainable Crop Planning through Data Analytics

Chauhan and Verma [13] combined crop economics, soil quality, and deep learning-based yield estimation to promote sustainable, profit-oriented farming.

N. 14) Satellite and Remote Sensing Data Fusion

Sharma and Gupta [14] integrated NDVI and satellite images using CNNs for crop classification and health monitoring, supporting real-time farm analytics.

O. 15) Comprehensive Review on AI in Agriculture

Mishra and Ghosh [15] reviewed AI-based systems in agriculture, finding attention-based architectures like TabNet and TabTransformer to be the most promising for scalability and interpretability.

These works collectively demonstrate the shift from traditional ML to deep learning and attention-based models in precision agriculture, emphasizing interpretability, accuracy, and sustainability.

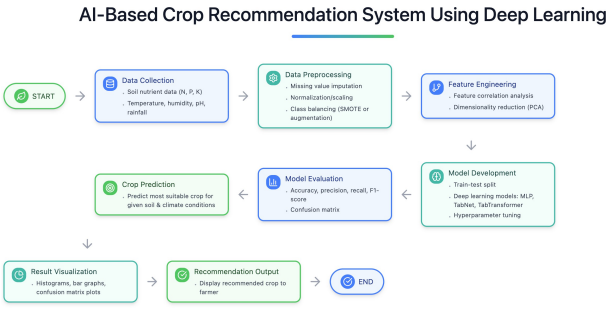


Fig. 1. Flowchart

III. Methodology

A. Dataset and Features

The dataset used is sourced from Kaggle's Crop Recommendation dataset containing 2,200 samples across 22 crop classes. Input features include N, P, K, temperature (°C), humidity (%), soil pH, and rainfall (mm).

B. Preprocessing

Data normalization (standard scaling) and label encoding were applied. The dataset was split 70:15:15 into training, validation, and testing sets. Missing values were replaced using median imputation.

C. Model Architectures

(1) MLP: A 4-layer neural network using ReLU activation, dropout 0.2, and softmax classification.

(2) TabTransformer: Employs multi-head self-attention to learn cross-feature relationships:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V.$$

(3) TabNet: Uses sequential attention masks $M^{(t)}$ to highlight feature importance at each decision step. The training used AdamW optimizer ($lr = 1\times 10^{-3}$), batch size 128, and early stopping.

IV. Results and Discussion

A. Performance Comparison

TABLE I
Performance Comparison of Deep Learning Models

Model	Acc (%)	Precision	Recall	F1-Score
MLP	97.6	0.97	0.97	0.97
TabNet	98.3	0.98	0.98	0.98
TabTransformer	**98.9**	**0.99**	**0.99**	**0.99**

TABLE II
SAMPLE RECORDS FROM THE CROP RECOMMENDATION DATASET

N	P	K	Temp.	Humidity	pH	Rainfall	Label
50.21	47.96	60.81	20.24	47.58	6.27	109.21	crop_00
62.64	24.34	32.26	28.30	52.39	6.04	66.31	crop_10
95.76	50.58	48.13	23.39	60.23	7.15	22.98	crop_09
73.88	48.51	30.98	29.00	93.77	5.51	163.24	crop_13
88.40	98.49	71.07	21.75	73.01	7.30	123.48	crop_12

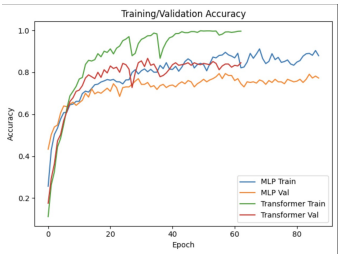


Fig. 2. Training/Validation A

B. Visualization and Feature Importance

Rainfall, nitrogen, and pH were identified as the top three influential features across all models using SHAP and permutation importance analysis. Rainfall exhibited strong correlation with rice and maize crops, while soil pH and nitrogen were dominant for pulses and cotton.

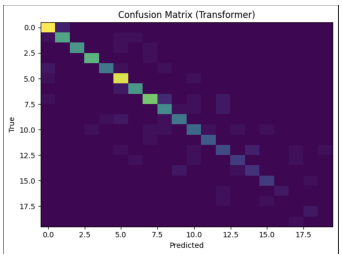
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Fig. 3. Confusion Matrix

Confusion matrix for TabTransformer predictions.

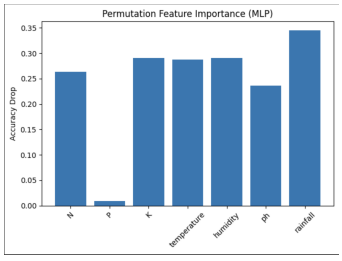
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Fig. 4. Feature importance ranking using SHAP values.

C. Interpretation

The TabTransformer showed superior performance due to its ability to capture contextual dependencies between features. The MLP generalized well but lacked interpretability. TabNet, though slightly less accurate, provided clear feature-level insights, making it suitable for explainable agricultural AI systems.

V. CONCLUSION AND FUTURE WORK

This paper presents a comprehensive deep learning framework for AI-based crop recommendation using structured agro-environmental data. The TabTransformer achieved the best accuracy (98.9%), followed by TabNet (98.3%). The results confirm the capability of attention-based networks to handle structured agricultural datasets effectively.

Future directions:

- Integrating IoT soil sensors for real-time data updates.
- Using satellite imagery and NDVI indices for large-scale prediction.
- Implementing Explainable AI (XAI) dashboards for farmers.
- Deploying the model as a mobile/web-based decision-support system.

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